

Paula Santás Rodríguez

paulasantas@outlook.com | (+1) 647 562 - 7381 | LinkedIn: PaulaSantasRodriguez

▼ Professional Summary.

I am a Mechanical Engineering graduate with minors in Biomedical and Robotics Engineering, focused on product development. Proven track record of delivering projects on time, on cost, and on spec while supporting cross-functional teams through DFMA-driven design optimization. Skilled in 3D modeling, prototyping, and iterative design using tools like SolidWorks and Onshape. Complemented by research experience designing precision jigs for biomedical experiments, demonstrating strong problem-solving and critical thinking abilities.

▼ Professional Experience

Product Development Engineer

Spin Master Ltd. | [August 2024 - Current]

- Leading the development of over **15+ medium to high complexity toy products** from concept to production, ensuring each is delivered **on time, on cost, and on spec** by actively managing timelines, cost, and technical feasibility.
- Created and revised complex 3D models and assemblies using **SolidWorks** and **Onshape**, supporting iterative design reviews, internal CAD Checkpoints, and tooling development with internal teams and overseas manufacturers.
- Applied **DFMA/DFM principles** to optimize injection- and blow-molded parts, reducing part complexity and achieving **average cost savings of 10% per unit** through material and tooling adjustments while still meeting spec.
- Collaborated with Asia, North America, and Europe-based vendors to resolve design-for-manufacturing issues, ensuring alignment between CAD intent and production capabilities while minimizing tooling delays.
- Drafted critical to consumer requirements, reviewed part quotations, and **helped the internal team aligned on feature priorities to support the sourcing team with price negotiations with the vendor**.
- Provided regular updates to executive stakeholders on design iterations, play pattern feasibility, cost status, and redesign timelines, directly influencing decision-making at cross-functional checkpoints.

R&D Mechanical Engineer

Sunnybrook Hospital Research Institute (FUS Lab) | [June 2023 – August 2024]

- Collaborated with **10+ researchers, and engineers** to define design goals and constraints for components used in **multiple experimental setups** supporting focused ultrasound therapies.
- Applied mechanical design expertise to create **tailored components** that **improved benchtop testing efficiency**, enabling faster transitions into clinical validation stages.
- Conducted independent research on medical and technical standards to deliver **100% compliant design outputs**, ensuring patient safety and material compatibility with MRI and clinical environments.
- Designed and prototyped complex components using **SolidWorks, 3D printing, and in-house machining tools (lathe, mill)**; produced detailed technical drawings and 3D files for third-party manufacturing of patient-centric moulds and parts.
- Conducted usability and safety testing across **10+ prototypes**, with **all final designs passing lab validation** and used in live research trials. Created documentation for the designed parts and for quality assurance and regulatory preparation, contributing to **clinical research studies progressing toward first-in-human trials**.

Engineering Innovation Intern

Spin Master Ltd. | [May 2022 – August 2022]

- Designed and prototyped proof-of-concept mechanisms using SolidWorks, Fusion 360, 3D printing, and foam molding techniques.
- **Achieved cost reductions** by minimizing material usage and combining mechanical functions in revised CAD files from Asia teams.
- Collaborated cross-functionally to prepare **presentation-ready prototypes** and support innovative, cost-efficient design solutions.

Product Development Engineering Intern

Spin Master Ltd. | [May 2021 – May 2022]

- **Managed 90+ projects** across two brands, delivering feasibility studies and early-stage cost estimates **within 10% accuracy**.
- Led design alignment meetings with global teams to resolve manufacturing challenges and reduce production delays.
- Supported **cost-saving initiatives** and product refinements presented to executives and integrated into company-wide planning.

▼ Skills

Experienced in Design (SolidWorks, OnShape), Modelling, Prototyping, and Programming (Python, C++, ROS, Git, Arduino, MATLAB).

Soft Skills in Problem solving, time management, project management, team leadership, and communication.

▼ Education

Bachelor of Applied Science and Engineering - University of Toronto

[September 2018 - Grad: May 2023]

- Major in **Mechanical Engineering** – Minors in **Biomedical and Mechatronics Engineering** and Certificate in **Engineering Business** from **UofT Rotman School of Management**